Predicting AC Optimal Power Flow Solutions Using Artificial Neural Network

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Abstract: Resolving Alternating Current Optimal Power Flow (ACOPF) problem is a vital task for grid operators to operate the grid safely and for the minimization of total generation cost. Power system networks are growing in size and complexity with time as a result of technology advancement, which raises demand and the need for Distributed Generations (DG). This causes demand and generation to become more unpredictable, necessitating the need to handle OPF problems more often than usual during the day, to meet the varying load with minimum cost. The AC-OPF problem can be solved by mathematical solutions; however, they need a significant computation time. With exceptional computational performance, artificial neural networks (ANN) are well-suited for function approximation applications. In this paper we employ a supervised learning approach to predict solutions for ACOPF problem by leveraging the general predicting capability of ANN. Hence, DNN (Deep Neural Networks) trained from mathematically solved data samples are used to predict ACOPF solutions. Predicted outputs are then processed from system knowledge perspective, and these results are used to compute power flow by enforcing reactive power limits by the solver. To validate the results, the approach is conducted on two IEEE test cases: IEEE 14 and IEEE 30 Test Bus systems using MATPOWER. Results are compared to interior point solver (IPS), and it is found that feasible solutions are generated with no optimality gap and voltage magnitude deviation of 0.023% from upper limit at the slack bus with in time scale of 1.62 times faster for IEEE 14-bus system. For IEEE 30 bus-system solutions are generated in 1.17x faster time scale and optimality gap of 0.02. For this system load bus voltage deviations from their upper limits are found only on two buses, to be 0.15% (for 59% of scenarios) at bus 25, and 0.27% deviation (for 8.8% scenarios) at bus 29.

Key – Words: *Optimal Power flow, artificial neural network, IPS, DG*

I. Introduction

The Optimal Power Flow (OPF) problem was first discussed by Carpentier in 1962 [1]. It is a very large and very difficult mathematical programming problem. It is solved every five minutes for the entire grid consisting of thousands of buses to get economical and reliable dispatch values [2] . The main goal of OPF is to reduce the costs of meeting the load demand for a power system while keeping up the security of the system.

Due to the slow convergence of iterative solvers, learning techniques to solve the OPF problem is becoming active area of research. Machine Learning techniques are potential solutions to solve variants of OPF problems with the goal of obtaining a more cost-effective solution and reducing the computational burden of solvers[3].

According to the type of studied OPF problems, [4] categorized available literature into: direct mapping of OPF variables [5], [6], [7], [8], [9], predicting active constraints [10], [11], mapping binary decision variables [12], learning control policy for OPF [13], [14], [15], [16], [17], stability constrained OPF [18], [19], [20], [21] and prediction of warm start points [22].

Direct prediction of OPF solutions with machine learning is the most popular approach in the literature. The idea is to leverage the universal approximation capability of DNNs (Deep Neural Networks) to learn the mapping between load input and OPF solutions [23]. A set of samples for input parameters of OPF is generated or gathered from historical data. These input parameters can be nodal power demand and renewable generation. The OPF solution corresponding to each sample point is obtained. A learner is trained with the input samples, given the OPF results (voltages, line flows, and power generation) as outputs [4]. In [23] such approach is validated on standard grids and evaluated feasibility on the proportion of predicted grids which satisfy all constraints and optimality on the ratio of average cost deviation of the predicted grid to the cost of the true grid. The main challenge of this approach is in ensuring the predicted ACOPF solutions to respect constraints. Constraint satisfaction corresponds to enforcing operational safety limits and adhering to physical laws, and is of paramount importance [11]. Feasibility is reported to be very poor at 51% for IEEE 30-bus system and 70% for IEEE 118-bus system with average cost deviation of 0.002 for both systems with a maximum of 10% load variations in the base load. Besides in the ACOPF problem formulation the authors didn't consider transmission line capacity limits which reduces the problem size in considering feasibility. To overcome the severe infeasibilities of directly predicting solutions [5] explored a method by parametrization of the ACOPF solutions to enforce generation limits. Strictly feasible solutions are used to train the learning model by generating OPF solutions from modified OPF problem called Restricted-ACOPF so that outputs are within limits in terms of voltage magnitude. This restriction helps to ensure predicted values to be feasible since the model learns from these sets. Prediction is implemented for all generator buses' voltage magnitude and active power generation except the slack bus. To recover active power generation of the slack bus and ensure feasibility of the overall AC OPF solution the power flow equations (PF) are solved. Even though high speedup factor (8x to 15x) and good optimality is claimed to be found, since the training sets are generated from modified problem, this makes the method less credible in representing typical ACOPF problem solution characteristic as the solution is often on the boundary of the feasibility set [24]. Besides the method didn't consider transmission line capacity constraint in the problem. The same method of predicting independent variables and reconstructing dependent variables of the ACOPF problem by solving PF equations is also employed in the work of [6]. A penalty approach is also employed in training the DNN so that the reconstructed solutions (generation, voltages, and branch flows) respect the corresponding operation limits. In [20] a method is proposed to create a scalable machine learning approach to predict ACOPF solutions that can reduce the significant amount of time required for training a neural network. The work utilizes spatial decomposition of the power network into a set of regions. In the first step the neural network learns to predict the flows and voltages on the buses and lines coupling the regions, and in the second step it trains, in parallel, the machine-learning models for each region.

Although using learners to predict OPF results directly may provide good-enough solutions, a slight error in predictions may result in suboptimal or infeasible solutions. In response to this challenge, predicting active constraint sets has also attracted attention[22]. This involves learning relevant sets of active constraints, from which the optimal solution can be obtained efficiently. And using the active sets as features allows us to reduce the dimension of the learning task [11]. Since Neural networks may learn solutions that violate physical constraints, and are thus untrustworthy in practical settings, [7] also explore optimal constraint prediction approach. In this setting, ANN model is trained to predict the set of constraints that are active in the optimal solution for some load distribution. A constraint is active if the corresponding state/control variable is at the maximum or minimum allowed value. The idea is to predict the active set of constraints by multi-label classification, and these values can be used to warm start other traditional optimization methods like interior point, so that time for computation is reduced by accelerating convergence. [10] employ classification algorithms from machine learning in order to learn the mapping from the uncertainty realization to the optimal active set for DCOPF.

In this thesis the direct approach by using Artificial Neural Networks for bus injection model of ACOPF will be assessed and will be evaluated on two standard IEEE test grids.

This paper is organized as follows: after brief overview of ACOPF and related works in Section I, Section II defines the problem mathematically. Section III presents the method used to solve ACOPF and explains how DNN training is done. In Section IV results are discussed and conclusion is made in Section V.

II. Problem formulation

The classical ACOPF problem is developed as follows:

$$min \sum_{i \, \varepsilon \, G} C_i(P_i) \qquad (1a)$$

Such that:

$$P_i(V,\delta) = P_i^G - P_i^L, \; \forall \; i \; \varepsilon \; N \qquad (1b)$$

$$Q_i(V,\delta) = Q_i^G - Q_i^L, \; \forall \; i \; \varepsilon \; N \qquad (1c)$$

$$P_i^{G\_min} \leq \; P_i^G \leq \; P_i^{G\_max}, \; \forall \; i \; \varepsilon \; G \qquad (1d)$$

$$Q_i^{G\_min} \leq \; Q_i^G \leq \; Q_i^{G\_max}, \; \forall \; i \; \varepsilon \; G \qquad (1e)$$

$$V_i^{min} \leq \; V_i \leq \; V_i^{max}, \; \forall \; i \; \varepsilon \; N \qquad (1f)$$

$$S_i \leq \; S_i^{max}, \; \forall \; i \; \varepsilon \; N \qquad (1g)$$

Where, $P_i \, and \, Q_i$ are net active and reactive power injected at bus i, P_i^G,Q_i^G,$P_i^L and Q_i^L$ are active and reactive power generated at bus i, and i-th bus demands respectively , S_i apparent power flow in lines in MVA, N number of nodes and G number of generators.

The objective (Equation (1a)) is minimization of total cost of generation where the cost of generation is given as a function of active power generated for each generator. Equations (1b) and (1c) captures the physics of the system while operational limits are enforced by Equations (1d) to (1g).

III. Methodology

The method we employ to learn a mapping between loading profiles and optimal solutions has four steps: ***a) Training:*** in the first step a DNN will be trained from historical or simulated data for a given topology of a power network. ***b) Predicting:*** using the trained DNN, outputs of the network which are control variables (active power generation and voltage magnitudes of generator buses) will be predicted for the reserved test set loading profiles (active and reactive power demands at load buses). ***c) Analyzing:*** as neural networks prediction might be outside of output limits, analyzing and correcting them from known knowledge is necessary. Therefore, predicted outputs which are below or above their operational limits will be treated as active constraints and the boundary value (lower or upper limit whichever is nearest) will be taken as primary solution. ***d) Processing:*** the final step is to compute power flow (PF) using these solutions as inputs by enforcing reactive power limits to be satisfied by the PF solver. This step in one hand enables one of the constraints to be satisfied (reactive power limits), on another hand it helps the line flows to be within limits. Once PF is solved, new control values; slack bus active power generation and voltage magnitudes for all generator buses will be generated. These processed values along with customized predicted values will be the final solutions. The approach is presented in block diagram below. (Figure 1).

Hence, load demands for n load buses (P_{Di},Q_{Di}; *for i=1 to n*) will be fed to the DNN and control variables for G generator buses (P_{Gi},V_{Gi}; *for i=1 to G*) will be predicted. Then these outputs of the network will be checked against their limits. If a value is predicted to be outside the limits, we will take it as active constraint and take the value at the boundary. To keep feasibility, the method employed is to enforce reactive power generation limits of generators by the solver while solving PF. The value of the slack bus active power generation (P_{G1}) and voltage magnitudes of PV buses (V_{Gi}, *for i=1 to G*) will be taken as new control values. Then results are reported by investigating whether solutions adhere to physical and engineering limits.

To measure optimality, we use average cost deviation in Equation (2a) which is average ratio of cost deviation in using predicted values to the true cost. To measure computation time gain (Equation (2b)) in using DNN we use average ratio of time ($t_{IPS,t}$) required to compute ACOPF by using MIPS (MATPOWER IPS) to the time spent in predicting solutions ($t_{ANN,t}$). This prediction time consists of the time required to predict and to solve PF. In literature [6], [7], [19] feasibility is reported in terms of how much of the test data used satisfy constraint, which is not able to show how far constraints are violated. But this paper shows how much bus voltages deviate from operational limits.

$$Avg. Cost\ Deviation \ = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{true}{}\right. \quad (2a)$$

$$Speedup\ Factor = \frac{1}{T}\sum_{t=1}^{T}\frac{t_{IPS,t}}{t_{ANN,t}} \quad (III\ b)$$

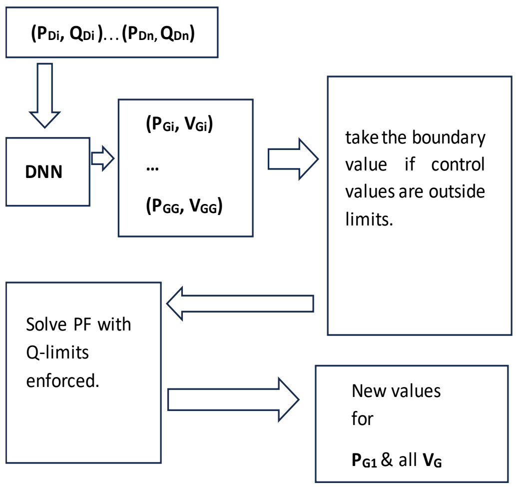

Figure 1. OPF solution Block Diagram

Generating Training Data

In supervised machine learning, labeled data points as many as possible are necessary to map features to target values and to increase the generalization capability of a model. On the other hand, huge data will take too much time for training. Therefore, it is critical to choose fair amount of data considering training time and processing capacity of processor used.

The training data required in our case is loading profiles (active and reactive power demands at load buses) and their optimal solution (active power generation dispatch and voltage magnitude at generator buses). Such data can be found from real grid operators or can be generated by simulation, in this paper we use simulation by perturbing MATPOWER7.1 [25] base loads for two standard buses (IEEE 14 and 30- bus systems). Since fixed active power to reactive power ratio is typical behavior for aggregated demand, we generate loading profiles by sampling the base loads randomly between 80% and 130% loading both for active and reactive power demand.

Hence, we generate 4,000 loading samples for IEEE 14-bus and 10,000 samples for IEEE 30-bus systems. These load profiles are then fed to MATPOWER7.1 Interior Point Solver, to solve ACOPF problem defined in equations (1a) to (1g) for IEEE 30-bus, but for IEEE 14-bus system equation (1g) is excluded as line limits are not included in the case file. The outputs of this solver are: optimal value of objective function (minimum total generation cost), optimal generators' active power dispatches, voltage magnitudes at generator buses and power flow solutions. Since it is required to map loading profile of the system to its solution, this data matrix is then used to train ANN and later used to predict solutions for different new load profiles. When these loading samples are solved for ACOPF problem, all samples for IEEE 14-bus system converged but only 4658 samples for IEEE 30- bus system converged and the remaining 5342 samples could not converge. This means either these loadings are unrealizable or cannot converge with the given iteration. In our training we use only converged samples.

Learning Model

A DNN model with default activation functions is used to learn a mapping between the load profiles and ACOPF solutions using Intel(R) Core (TM) i5-7200U CPU @ 2.50GHz, 4GB RAM Processor. The size of the input layer for input features is equivalent to two times the number of load buses (for P_D and Q_D) and that of output layer is two times the number of PV buses (for P_G and V_G). To find optimal size of network hyper parameters: number of hidden layers, size of hidden layers, number of epochs and type of training algorithm; we employ grid search using MATLAB13a by searching least possible RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for each model. We used data division of 70% for training and 15% each for validation and testing. Regardless of dimensionality we didn't exclude the slack bus P_{G1} from the training data, intuitionally expecting its relevance in learning the natural relationships between loading profiles and output generations and voltage magnitudes. It is found that the final architecture to be used for IEEE-30 bus system is, using Levenberg Marquardt training algorithm, three hidden layers with 40, 20 and 20 neurons in consecutive layers, for 400 epochs, and for IEEE-14 bus system is, using Levenberg Marquardt training algorithm, three hidden layers with 15,10 and 10 neurons in consecutive layers, for 400 epochs.

IV. Results and discussions

For IEEE 14-bus system test prediction is done on 600 samples of test set. It is found that all predicted voltage magnitudes are in range, and all predicted active power generations (P_G) are in range but as shown in Table 1 some samples of predicted active power generations at three buses , violate lower limits for active power generation So, they are considered to be active constraints and treated as such by taking the lower limit to be their value at 0 MW.

Table 1 Samples of predicted outputs which violate constraints

IEEE System	Bus number	% of Samples violated constraint limits	
		P_G	$\|V_G\|$
14 Bus	2	2.8%	-
	6	19.6%	-
	8	10.3%	-
30 Bus	2	-	1.14%

Then these modified values are used to compute power flow with reactive power limits enforced by the solver, so as to construct slack bus power generation value and to check feasibility of predicted control variables. It is found that solutions to all control variables are feasible and average slack bus voltage is 1.060261856 with deviation of 0.000236PU (0.023%) from upper limit of 1.06PU (Table 2). With these predicted active power generations of P_G at buses 2,3,6,8 for PV buses and calculated slack bus P_{G1}, average total cost of generation becomes 8508.9$/hr., which is 8509.04 $/hr. when calculated in MIPS. When calculated using MIPS, the average time required for computing ACOPF for the test set is 0.057488 sec., and when computing is done using ANN the average time spent including prediction time and time for PF computation becomes 0.026598333 sec. + 0.010749 sec.= 0.037347333 sec. Even though the cost improvement is not significant, time for computation is improved by a speed up factor (Equation ((IIIb) of 1.62x.

For IEEE 30-bus system prediction is done on 699 samples of test set. Following same procedure, it is found that all predicted active power generations are in range, and all voltage magnitudes are in range except at generator bus 2, where 1.14% of samples (Table 1) are little lower than 0.95PU with average value of 0.940779387PU. So, it is considered as active constraint by taking the lower limit as its value at 0.95PU. Then this modified value is used to compute power flow with Q limits enforced by the solver and reconstruct slack bus power generation. It is found that, as shown in Table 2 solutions to all control variables are feasible, but 59% of predicted test samples violate maximum voltage magnitude limit at load bus-25 (i.e., V_{25}) by average voltage deviation factor of 0.0015 PU (0.15%) from 1.05PU and 8.8% of samples at load bus-29 violate maximum voltage magnitude limit (i.e., V_{29}) by average voltage deviation factor of 0.0027(0.27%). With these predicted active power generations and voltage magnitudes for PV buses and calculated slack bus active power generation, average total cost of generation becomes 529.47$/hr. while this cost is 515.37$/hr. when calculated using MIPS. The average time required for computing ACOPF for the test set using MIPS is 0.0902 sec. and using ANN the average time including prediction time and time for PF computation becomes 0.0727sec.+0.0071sec.= 0.0798 sec. This shows that even though no cost improvement is made, with optimality gap of 0.02, time for computation is improved with speed up factor of 1.17x.

Table 2 Voltage magnitude deviations from their upper limit

IEEE System	Bus number	%$\|V_G\|$ deviation	% of test sets with $\|V_G\|$ deviation
14 Bus	1	0.023%	100%
	25	0.15%	59%
30 Bus	29	0.27%	8.8%

V. Conclusions

In this work, we presented a new approach that would leverage machine learning for solving ACOPF. It is shown that feasible solutions can be generated from ANN predictions in a faster time scale, by processing predicted outputs through solving power flow and enforcing reactive power limits by the solver. Even though constraint violations cannot be avoided perfectly, it is shown that the value of violations can be reduced to a significant amount with considerable computation time speed up gain, while the cost of operation is nearly equivalent to standard interior point Solver. Besides the amount of constraint violation in terms of voltage deviation is measured. Future work will be experimenting to reduce this voltage deviation and evaluating on more complex grids using GPU technology to increase computation speed up.

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